

TAKUA PA, Thailand

WMO: 485610

Lat: 8.6842N

Lon: 98.2522E

Elev: 25

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	70.7	72.1	64.4	90.6	79.8	66.3	97.0	79.4	12.2	80.0	10.8	80.7	0.1	N/A	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	13.7	93.6	80.8	92.3	80.2	91.2	79.9	83.1	91.0	82.4	89.9	81.7	88.9	2.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.7	88.4	80.2	156.9	87.3	79.7	154.3	86.4	47.0	91.4	46.2	90.2	45.5	89.3	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.7	8.5	68.1	96.0	1.6	1.7	66.9	97.2	65.9	98.2	65.0	99.2	63.8	100.5
			65.2	84.0	2.2	1.5	63.6	85.1	62.3	86.0	61.1	86.9	59.5	88.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.8	80.9	81.4	82.6	83.7	83.0	82.4	82.0	81.7	80.9	80.7	81.2	80.9
	DBStd	2.04	1.52	1.51	1.87	1.82	2.07	2.10	2.08	2.27	1.80	1.60	1.32	1.59
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11602	957	880	1011	1011	1024	971	992	982	927	952	937	958
	CDD65	6127	492	460	546	561	559	521	527	517	477	487	487	493
	CDH74	63245	4897	4903	6091	6423	6210	5648	5612	5331	4487	4350	4556	4737
	CDH80	22978	1866	2059	2667	2750	2324	1992	1885	1690	1271	1247	1543	1684
(14) Wind	WSAvg	1.6	1.5	1.4	1.3	1.1	1.5	2.1	2.4	2.3	1.8	1.2	1.0	1.5
(15) Precipitation	PrecAvg	144.6	1.9	1.6	5.5	8.2	15.9	16.7	17.4	21.0	24.0	19.5	9.5	2.7
	PrecMax	216.1	8.6	4.9	22.4	21.4	31.9	32.1	35.4	39.6	41.6	40.7	23.5	9.4
	PrecMin	99.5	0.0	0.1	0.3	0.7	6.7	5.7	8.1	9.1	11.3	8.8	3.4	0.5
	PrecStd	22.2	2.0	1.2	4.7	4.4	5.9	5.8	5.0	6.5	7.8	6.9	5.1	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.4	93.8	95.2	95.8	94.8	92.3	91.8	91.6	90.4	90.1	90.9	91.5
		MCWB	76.4	78.2	80.6	82.8	82.5	82.0	82.3	82.9	81.3	79.7	79.4	77.2
	2%	DB	90.6	91.9	93.0	93.8	91.9	90.3	89.8	89.5	88.8	88.5	89.6	90.0
		MCWB	76.8	77.9	80.0	81.9	81.3	81.0	81.2	80.3	79.5	78.9	77.5	77.5
	5%	DB	89.2	90.3	91.5	91.8	90.3	89.0	88.3	88.0	87.5	87.4	88.5	88.7
		MCWB	76.8	77.5	79.4	81.1	81.3	80.7	80.2	80.0	79.7	79.3	78.7	77.5
	10%	DB	87.8	88.9	90.1	90.4	88.8	87.9	87.1	86.9	86.2	86.2	87.4	87.4
		MCWB	76.6	77.2	78.9	80.6	80.8	80.2	79.8	79.7	79.4	79.1	78.5	77.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.1	80.9	82.9	83.9	83.7	83.3	83.5	83.4	82.4	81.6	81.2	80.9
		MCDB	88.3	90.3	92.2	93.8	92.0	90.3	90.5	90.5	89.1	88.0	88.3	88.0
	2%	WB	79.2	79.9	81.4	82.7	82.6	82.1	81.9	82.1	81.0	80.6	80.4	79.8
		MCDB	87.0	88.8	90.5	91.8	90.2	88.7	88.3	88.6	87.1	86.6	87.2	86.7
	5%	WB	78.5	79.1	80.5	81.9	81.8	81.4	80.9	81.0	80.3	79.9	79.8	79.1
		MCDB	86.1	87.5	89.3	90.3	88.8	87.6	86.7	86.6	85.9	85.6	86.4	85.9
	10%	WB	77.9	78.3	79.8	81.1	81.1	80.7	80.2	80.1	79.7	79.3	79.1	78.3
		MCDB	85.2	86.4	88.0	88.8	87.6	86.5	85.5	85.2	84.9	84.7	85.5	85.0
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	16.6	15.5	13.7	11.3	10.2	9.7	9.5	10.2	10.9	12.8	13.9
		MCDBR	17.2	18.6	17.8	16.1	13.7	12.3	11.8	11.3	12.3	13.0	14.6	15.6
		MCWBR	6.7	7.4	7.0	6.3	5.7	5.1	5.0	4.8	5.3	5.6	6.0	6.3
	5% WB	MCDBR	15.3	16.8	15.9	14.8	12.6	11.4	10.7	10.7	11.2	12.0	13.6	14.2
		MCWBR	6.6	7.3	6.7	6.3	5.7	5.4	5.2	5.2	5.3	5.5	6.1	6.4
(40) Clear-Sky Solar Irradiance	taub	0.463	0.484	0.499	0.482	0.457	0.461	0.463	0.453	0.436	0.444	0.453	0.463	(40)
	taud	2.293	2.211	2.211	2.264	2.379	2.386	2.361	2.385	2.427	2.397	2.356	2.306	(41)
	Ebn at Noon	270	268	264	265	266	262	263	269	277	276	271	267	(42)
	Edh at Noon	42	47	48	45	39	38	39	39	38	39	39	41	(43)
(44) All-Sky Solar Radiation	RadAvg	1657	1916	1932	1867	1532	1475	1454	1485	1472	1446	1479	1467	(44)
	RadStd	122	109	171	117	120	107	95	106	110	146	108	127	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	+0.29	+0.56	N/A	N/A	+0.32	+0.38	N/A	N/A	N/A	N/A

Nomenclature: See separate page